

Q2: Bertrand's Paradox

Problem Statement

A chord of a circle is chosen "at random." What is the probability that its length exceeds the side length of the inscribed equilateral triangle?

Show that different mathematically natural models of "random chord" lead to different answers.

Solution

Preliminaries

Consider a circle of radius R (take $R = 1$ for simplicity, as the probability is scale-invariant).

The side length of an inscribed equilateral triangle in a circle of radius R is:

$$s = \sqrt{3}R.$$

For a chord with central angle θ ($0 \leq \theta \leq \pi$), the chord length is:

$$L = 2R \sin\left(\frac{\theta}{2}\right).$$

We want $L > s = \sqrt{3}R$:

$$2R \sin\left(\frac{\theta}{2}\right) > \sqrt{3}R$$

$$\sin\left(\frac{\theta}{2}\right) > \frac{\sqrt{3}}{2}$$

$$\frac{\theta}{2} > \frac{\pi}{3} \quad (\text{since } \theta/2 \leq \pi/2 \text{ for chords with } \theta \leq \pi)$$

$$\theta > \frac{2\pi}{3}.$$

Thus the chord is longer than the triangle side if and only if the central angle $\theta \in (2\pi/3, \pi]$.

Model 1: Random endpoints on circumference

Choose two points independently and uniformly on the circumference. The central angle θ is uniformly distributed on $[0, \pi]$ (taking the smaller angle between the two points).

Probability:

$$P_1 = \frac{\pi - 2\pi/3}{\pi} = \frac{\pi/3}{\pi} = \frac{1}{3}.$$

Model 2: Random distance from center (random radius, perpendicular chord)

Choose a random distance d from the center to the chord, uniformly in $[0, R]$, and a random direction uniformly in $[0, 2\pi)$. The chord is perpendicular to the radius at that distance.

Chord length:

$$L = 2\sqrt{R^2 - d^2}.$$

Inequality $L > \sqrt{3}R$:

$$2\sqrt{R^2 - d^2} > \sqrt{3}R$$

$$\begin{aligned}
4(R^2 - d^2) &> 3R^2 \\
4R^2 - 4d^2 &> 3R^2 \\
R^2 &> 4d^2 \\
d &< \frac{R}{2}.
\end{aligned}$$

Since d is uniform on $[0, R]$:

$$P_2 = \frac{R/2}{R} = \frac{1}{2}.$$

Model 3: Random midpoint of chord inside circle

Choose the midpoint M of the chord uniformly within the disk (uniform by area). Let r_M be the distance of M from the center.

Chord length:

$$L = 2\sqrt{R^2 - r_M^2}.$$

Inequality $L > \sqrt{3}R$:

$$\begin{aligned}
2\sqrt{R^2 - r_M^2} &> \sqrt{3}R \\
\sqrt{R^2 - r_M^2} &> \frac{\sqrt{3}}{2}R \\
R^2 - r_M^2 &> \frac{3}{4}R^2 \\
r_M^2 &< \frac{1}{4}R^2 \\
r_M &< \frac{R}{2}.
\end{aligned}$$

This means the midpoint lies inside a disk of radius $R/2$. The area of this smaller disk is $\pi(R/2)^2 = \pi R^2/4$. The total area is πR^2 .

Since the midpoint is uniform over area:

$$P_3 = \frac{\pi R^2/4}{\pi R^2} = \frac{1}{4}.$$

Summary of Results

$$\boxed{P_1 = \frac{1}{3}} \quad (\text{random endpoints})$$

$$\boxed{P_2 = \frac{1}{2}} \quad (\text{random distance from center})$$

$$\boxed{P_3 = \frac{1}{4}} \quad (\text{random midpoint by area})$$

Why the Answers Differ

Bertrand's paradox illustrates that the phrase "random chord" is ambiguous. Each of the three methods describes a different natural way to choose a chord uniformly:

- **Method 1 (endpoints):** Uniformly picks two points on the circumference. This corresponds to a distribution where chords are more likely to be longer because endpoints spread out.
- **Method 2 (radial distance):** Uniformly picks the perpendicular distance from the center. This weights all distances equally, leading to a different distribution of chord lengths.
- **Method 3 (midpoint):** Uniformly picks the midpoint by area inside the circle. This favors midpoints near the center (since area increases with radius), making shorter chords more likely.

The three methods are all "uniform" in different geometric senses, demonstrating that probability problems must specify the sampling mechanism precisely to obtain a unique answer.